OXYPC INVENTORY LIFECYCLE

PRODUCT

Product Overview

Why the app exists, where it is going, how success is measured, and the key findings to fix first.

Last updated: 6 Nov 2026 · Source: staging branch @ 55ce58a

App Development Team · Bluemonk

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In this document: Part A — Key Findings · Part B — Product Vision · Part C — Roadmap · Part D — KPI Definitions

PART A

Key Findings

Writing these documents meant reading the code end to end. Along the way the review found gaps between what the business expects and what the code does today. They are listed here so they can be triaged; each links to the document with the detail and the file references.

Fix first: security

#	Finding	Why it matters	Detail
S1	Uploaded files are served at `/uploads` with no login check (`main.py` mounts `StaticFiles`)	KYC documents, payment proofs, GRN invoices and daily stock files can be downloaded by anyone on the network who knows or guesses a path	Security, Integrations
S2	Page permissions default to "allowed" for custom roles	A new custom role (including the auto-created `sub_admin`) can open every page that is not admin-only until someone restricts it	Security
S3	Default secrets ship in code and seed scripts	DB password, OxyQC agent key, admin and UAT passwords are known defaults unless changed on the server	Security
S4	Permission changes and admin password resets are not written to the audit log	No record of who widened someone's access	Security
S5	CSV exports include prices even when Pricing Visibility is off	Pricing leaks through exports	Permission Matrix
S6	Pricing visible to every role by default (`can_view_pricing` defaults on; P&L gate admits non-management and custom roles). Policy of 1 Nov 2026 is management-only	Buying cost, margin and P&L exposed to floor and sales staff. Fix: default Pricing Visibility off, restrict the P&L gate, strip prices from exports, default-off Cost Config, Price Matrix and Product Pricing	Security (Pricing and financial data policy, R13)

Fix first: data safety

#	Finding	Why it matters	Detail
D1	App, database and backups all sit on one server; `uploads/` is not backed up	One disk or server failure loses data and files	Runbook & DR
D2	Database dumps are made without `--clean` , but the restore script expects to replace tables; no test restore is on record	A restore may fail when it is needed	Runbook & DR
D3	No recovery targets (RPO / RTO) are set	Nobody has agreed how much data loss or downtime is acceptable	NFRs

Process rules the code does not enforce

#	Finding	Detail
P1	A tag can be sold from any stage, not only Ready for Sale. This is deliberate (director's decision, 19 Oct 2026, `services/control_engine.py::validate_sale_allowed`); only blocked tags are refused. Confirm it is still wanted, as the Sales role is described as selling Ready for Sale stock	State Machine
P2	Several stage moves accept any signed-in user and do not check the tag's current stage (L1/L2 Mark Complete, Stress Test fail/complete, Back to Inventory)	State Machine
P3	The three lists of stages and allowed moves in the code disagree with each other	State Machine, Data Model
P4	Grades differ by screen: Final QC offers A/B/C, the device model A–E plus scrap, CRM A/B/C/D/Mix/As-Is/Scrap	PRD-03
P5	The `returned` stage is never used; a return moves the tag straight to Stock In	State Machine

Roles that do not exist in the code yet

Production Manager, Final QC Inspector, Accounts and TRC Manager have no built-in role. Production work runs under `inventory_manager`, and the Accounts pages refuse a custom Accounts role because `routers/accounts.py` checks a fixed list (`FINANCE_ROLES`). Fifteen templates still check role names directly instead of using `role_allowed()`. See Permission Matrix.

Features expected but not built

#	Expected	Today	Detail
F1	E-way bill generation	Only e-way bill numbers/files for goods coming in are stored	PRD-04, Integrations
F2	GST invoice generation	A printed HTML tax invoice, one tag per invoice, HSN fixed at 8471, GST in floating point, free-text invoice number	PRD-04
F3	Rentals	No code found	Roadmap
F4	Staging server	Auto-deploy pushes `main` straight to production	Deployment
F5	Automated test run	172 test files, no CI job runs them	NFRs

Housekeeping

- `docs/schema.dbml` (29 Apr 2026) is missing 58 of the 147 tables in `models/`; the approved diagram was never saved back. Data Model
- WhatsApp was removed, but `services/care_service.py` still calls it, so Care offer sends fail. Integrations
- `.github/workflows/deploy.yml` still restarts the pre-Docker service. Deployment
- Dashboard: "Revenue This Month" ignores the overrides "Profit This Month" uses; "Active lots" counts trashed lots; the P&L date filter changes nothing on screen. KPI Definitions
- The `CLAUDE.md` note on `is_trashed` is out of date: the columns are now NOT NULL. ADRs

PART B

Product Vision

Scope of this document: why the app exists, who it serves, what "good" looks like, and what is in and out of scope. Every statement about current behaviour is taken from the code on branch `staging-docs`. Anything not in the code yet is marked **Proposed**.

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1. The problem

OxyPC Computers / Bluemonk buys used laptops, desktops and TFT monitors in lots, refurbishes them and sells them. One device passes through up to ten teams before it is sold:

```
Buy lot → GRN → IQC → Stock In → (Production) → L1/L2 or L3/L4 Repair → Stress Test (QC)
       → Cosmetic (Cleaning, Putty, Dry Sanding, Masking, Painting, Water Sanding)
       → Final QC (grade) → Ready to Sale → Sold → (Return / Credit Note / Replacement)
```

The stages are real code, not a diagram: `models/device.py::DeviceStage` has 24 stages, and every move is checked against the `AllowedTransition` table by `services/control_engine.py::validate_transition`.

What still hurts, as seen in the code and the agreed opportunities:

#	Pain	Evidence in code today
P1	Engineers cannot see where their part request stands, and "Part Not Available" (PNA) is tracked apart from the request.	`models/part_request.py::PartRequest.status` and `models/pna_part.py::DevicePNAPart` are separate tables with no link. Status is shown only on Device Detail (`templates/devices/detail.html`). No notification on handover (`routers/part_requests.py`).
P2	Fail reasons do not travel with the tag. Each fail overwrites the last one.	`Device.stress_notes` is cleared on Complete; `Device.repair_notes` is overwritten by a Cosmetic fail; `Device.fqc_failure_reason` keeps one value (`models/device.py` , `routers/cosmetic.py::cosmetic_fail_assign`). History lives only as free text in `StageMovement.notes` .
P3	The cosmetic chain moves one tag at a time.	`routers/cosmetic.py::advance_stage` takes one barcode. `bulk_assign` only reassigns (it advances only Cosmetic Received → Cleaning).
P4	Grade does not drive the price.	Final QC sets `Device.grade` (A/B/C in `templates/cosmetic/final_qc.html`). Ready to Sale min price = unit cost + other rate + margin (`services/ready_pricing.py::tag_min_price`) — grade is not an input. The CRM `GradePriceMatrix` (`models/crm.py`) is not used outside `routers/crm_price_matrix.py` .
P5	Harvested parts are not used first.	Part Master keeps New and Harvest rows apart (`routers/spare_parts.py::parts_list` , `SparePart.source`). Handover uses whatever `part_id` the request names (`routers/part_requests.py::handover_part`).
P6	Invoice and e-way bill are not produced from the sale.	Invoice is a per-device HTML print (`routers/invoices.py::print_invoice`), invoice number is typed or parsed from an uploaded PDF (`routers/sales.py::parse_invoice_pdf`). E-way bill exists only on inbound lots / GRN / Parts GRN (`models/lot.py::e_way_bill` , `models/parts_grn.py::eway_bill_*`), never on a sale.

#	Pain	Evidence in code today
P7	No single TRC view.	The TRC Manager has to open Production Manager (`/trc-production`), Part Master (`/spare-parts`), WorkID Status (`/workid-status`), Daily Stock (`/reports/daily-stock`) and Physical Audits (`/audits`) separately.

2. Vision statement

Every device, every hour: one tag, one truth. Anyone at OxyPC can scan a tag and see where it is, why it is there, what it is waiting for, what it cost and what it should sell for — and the next person in the chain is told without being asked.

3. Target users — the 12 personas

App role keys come from `models/user.py::UserRole` and `ROLE_LABELS`. Roles are free text, so custom roles (for example `trc_manager`) are allowed and governed by the Module Permission matrix (`auth/dependencies.py::require_roles` , `models/role_permissions.py`).

#	Persona	App role (today)	Main job in the app	Main pages
1	IQC Inspector	`iqc_inspector` ("IQC Handler")	Inspect incoming devices, record specs and parts required	`/iqc` , OxyQC agent
2	Inventory Manager	`inventory_manager`	Lots, GRN, Stock In, stage moves, flags, credit-note table	`/lots` , `/grn` , `/devices` , `/transfers`
3	Production Manager	custom role / matrix (`production_manager` module)	Allocate tags to engineers, buckets, PNA summary	`/trc-production`
4	Repair Engineer L1/L2, L3/L4	`l1_engineer` , `l2_engineer` (legacy), `l3_engineer`	Repair, raise part requests, mark PNA, hand to Stress	`/repair/l1` , `/repair/l3l4` , Device Detail
5	QC Inspector (Stress Test, OS install, diagnose)	`qc_inspector` ("QC Handler")	Stress test, pass to Cosmetic or fail to L1/L2	`/stress` , `/qc`
6	Cosmetic Lead	`cosmetic_manager`	Run Cleaning → Putty → Dry Sanding → Masking → Painting → Water Sanding	`/cosmetic/*`
7	Final QC Inspector	permission `cosmetic_finalqc`	Manual diagnose, grade, pass to Ready or fail back by reason	`/cosmetic/final_qc`
8	Spare Parts Manager	`spare_parts_manager` ("Parts Manager")	Hand over, Not In Stock, Procure, Parts GRN (As New), Harvest	`/spare-parts` , `/parts-grn`
9	Sales Executive	`sales` , `sales_manager` , `telecaller`	Ready for Sale, block tags, sell, challan, gate pass, returns	`/sales/ready` , `/block-tags` , `/sales` , `/gate-pass` , `/returns`
10	Accounts	custom role / matrix	Payments, invoice, sales list, daily stock	`/accounts/*` , `/sales` , `/reports/daily-stock`
11	Managers / Admin	`admin` , `sales_manager`	Dashboard, reports, P&L, user tickets	`/dashboard` , `/reports/*` , `/tickets`

#	Persona	App role (today)	Main job in the app	Main pages
12	TRC Manager	custom `trc_manager`	Oversee inventory, parts, production; warranty extension; daily stock; physical audits	`/trc-production`, `/spare-parts`, `/extended-warranty`, `/audits`

Note: Rentals are listed as a Sales Executive task, but no rental feature was found in `routers/`, `models/` or `templates/`. See Open questions.

4. Value proposition

For	Who need to	The app gives them	Unlike
Shop-floor teams (IQC, Repair, QC, Cosmetic, Final QC)	Know what to work on next and why	A role queue per stage with WorkIDs, tag history and (proposed) the full fail and part story on the tag	Paper job cards and WhatsApp messages
Spare Parts	Fill requests fast and use stock we already own	Request → Handover → Verify flow, Parts GRN, Harvest stock; (proposed) PNA status and harvest-first	Separate spreadsheets for new and harvested parts
Sales and Accounts	Sell at the right price and bill correctly	Ready-to-Sale pricing, block tags, sale with GST invoice, challan, gate pass, credit notes; (proposed) grade price and e-way bill from the sale	Re-keying sale data into a billing tool
Managers, TRC Manager	See flow, cost and profit	Overall Dashboard, Business P&L, Lot P&L, Daily Stock, Stock Aging; (proposed) one TRC view	End-of-day calls to each team lead

5. Goals and success measures

Targets are measurable. Where the app does not yet store or show the figure, the baseline is "to be measured". Targets marked *(proposed)* need sign-off from the owner named in 05_KPI_Definitions.md.

Goal	Success measure	Baseline	Target	By
G1 Faster flow	Median days from Stock In to Ready to Sale (from `stage_movements`)	baseline: to be measured	−20% vs baseline *(proposed)*	Q2 2027
G1 Faster flow	Tags older than 30 days in any one stage (stage ageing)	baseline: to be measured	−50% vs baseline *(proposed)*	Q2 2027
G2 Parts never block silently	Median PNA wait time (mark → clear, `device_pna_parts`)	baseline: to be measured	≤ 5 working days *(proposed)*	Q1 2027
G2 Parts never block silently	Parts fill rate: requests handed over within 1 working day	baseline: to be measured	≥ 85% *(proposed)*	Q1 2027
G2 Parts never block silently	Share of handed-over parts that came from Harvest stock	baseline: to be measured	+25% vs baseline *(proposed)*	Q2 2027
G3 Right first time	First-pass yield at Stress Test and at Final QC	baseline: to be measured	≥ 85% at Final QC *(proposed)*	Q2 2027
G3 Right first time	Tags failed at Final QC with a fail reason that is still visible at the next stage	0% for Stress notes after Complete (cleared in code)	100%	Q4 2026

Goal	Success measure	Baseline	Target	By
G4 Price by grade	Ready-to-Sale tags with a grade-based suggested price	0% (grade not used in `services/ready_pricing.py`)	100% of graded tags	Q1 2027
G5 Clean billing	Sales with an invoice number recorded	baseline: to be measured (`sales.invoice_no` null rate)	100%	Q1 2027
G5 Clean billing	Sales above the e-way bill threshold with an e-way bill number	0% (no field on `sales`)	100%	Q2 2027
G6 Trust the numbers	Tags at `sold` with no Sale row ("orphans", `routers/reports.py::_sold_orphan_rows`)	baseline: to be measured	0	Q4 2026
G6 Trust the numbers	Write operations covered by the audit log	~35% (`docs/audits/2026-04-29-claude-md-full-audit.md`, Section 5)	≥ 90%	Q2 2027
G7 Fast pages	Dashboard loads over 1 s (logged by `routers/dashboard.py::dashboard`)	baseline: to be measured from server log	< 5% of loads	Q4 2026
G8 Adoption	Personas with signed UAT for their journey (06)	0 of 12 for this release	12 of 12	Each release

6. Scope

In scope (this vision, next three quarters)

Area	In scope
Device lifecycle	GRN → IQC → Repair → Stress → Cosmetic → Final QC → Ready → Sold → Returns
Parts	Part requests, PNA, Parts GRN (As New), Harvest, warranty part cycle
Sales	Ready to Sale pricing, block tags, sale, invoice, delivery challan, gate pass, returns, credit notes, warranty extension
Management	Overall Dashboard, Business P&L, Lot P&L, Daily Stock, Stock Aging, Receivables, Overdue, Physical Audits, Tickets
Proposals	The seven agreed opportunities (PRD-01 to PRD-04 plus batch cosmetic moves, harvest-first, one TRC view)

Out of scope (for now)

Area	Why
CRM sourcing / telecalling / dealer calling redesign	Separate product line in the same app; not part of the agreed opportunities
Reliance FieldOps (`/fieldops`)	Standalone offline app; touches no OxyPC tables (`README.md`)
Trade Partner portal and external partner bidding	Not part of the refurbishment flow being improved
Full accounting ledger (GL, TDS, bank reconciliation)	Accounts uses an external tool; the app records payments and invoices only
Multi-tenant SaaS	Mentioned in BRD v6 objectives; not in this horizon

Area	Why
Rentals	Not found in code; needs discovery first (see Open questions)

7. Guiding principles

1. **The tag is the unit of truth.** Every fact about a device hangs off its tag and stays in its history. Never overwrite a fact that explains a delay.
2. **Server decides.** Stage moves go through `validate_transition`; prices and totals are computed server-side in `Decimal` (project skill `server-side-financial-snapshot`).
3. **Tell the next person.** When work changes hands, the receiver gets a queue entry and a notification (`services/notifications.py`), not a phone call.
4. **One number, one source.** A tile and its drill-down use the same query (the pattern in `routers/stock.py::_prod_tile_criteria` and `services/business_pl.py`).
5. **Permissions from the matrix.** Gate buttons with `role_allowed()` / the Module Permission matrix, never hard-coded role lists (project skill `custom-role-permissions`).
6. **Same look everywhere.** New screens use the Design System: summary tiles → filter bar → alert → tabs → Global Table card; success colour `#0F7A37` (`qa_docs/design_system/design_system.html`).
7. **Audit before change.** Every write that matters is in `audit_logs` (`services/audit_engine.py::audit`); no schema change without dbdiagram approval (`CLAUDE.md`).

8. Risks

#	Risk	Likelihood	Impact	Mitigation
R1	Schema changes needed by PRDs wait on dbdiagram approval	Medium	Delays Now items	Batch the schema asks into one review per quarter; keep changes additive (`db_validator.py` auto-provisions additive columns)
R2	No dedicated staging server exists in code; autodeploy goes straight from `origin/main` to production (`deploy/oxypc-autodeploy.sh`)	High	Untested changes reach users	Stand up staging before the first UAT cycle (06); deploy only after sign-off
R3	Dashboard numbers drift from report numbers	Medium	Managers lose trust	Shared computations (`services/business_pl.py`); KPI owners sign 05
R4	Grade-based price makes Sales feel second-guessed	Medium	Low adoption of PRD-03	Suggest, do not block; keep manual override with reason
R5	E-way bill / e-invoice rules change or need a GSP account	Medium	PRD-04 rework	Phase 1 captures numbers only; API integration only after Accounts confirms provider
R6	Test coverage is thin for business rules (audit Apr-29: testing 3.0/10)	High	Regressions	Regression checklist in 06; grow `tests/` (173 files today) per PRD
R7	Free-text roles mean a new custom role silently lacks a module	Medium	Blocked users at go-live	UAT entry criterion: permission matrix reviewed per persona
R8	Audit trail covers ~35% of writes (Apr-29 audit)	High	Disputes cannot be traced	Every PRD adds audit actions for its writes

9. Open questions

#	Question	Owner
Q1	Where are rentals handled today? No code found.	Sales lead
Q2	Is the Production Manager a custom role in production, or does `inventory_manager` do this job?	TRC Manager
Q3	Who is Accounts in the app (custom role name)?	Admin
Q4	Do we need e-invoice (IRN) as well as e-way bill?	Accounts
Q5	Grade scale: Final QC offers A/B/C, `DeviceGrade` has A–E + scrap, CRM uses A/B/C/D/Mix/As-Is/Scrap. Which is the sales grade?	Final QC lead + Sales lead

Sources read

`CLAUDE.md`, `TASKS.md`, `README.md`, `docs/audits/2026-04-28-as-is-audit.md`, `docs/audits/2026-04-29-claude-md-full-audit.md`, `docs/generate_brd_v6.js`, `models/device.py`, `models/user.py`, `models/part_request.py`, `models/pna_part.py`, `models/crm.py`, `models/sales.py`, `routers/part_requests.py`, `routers/cosmetic.py`, `routers/stress_api.py`, `routers/sales.py`, `routers/invoices.py`, `routers/spare_parts.py`, `services/ready_pricing.py`, `deploy/oxypc-autodeploy.sh`, `qa_docs/design_system/design_system.html`.

PART C

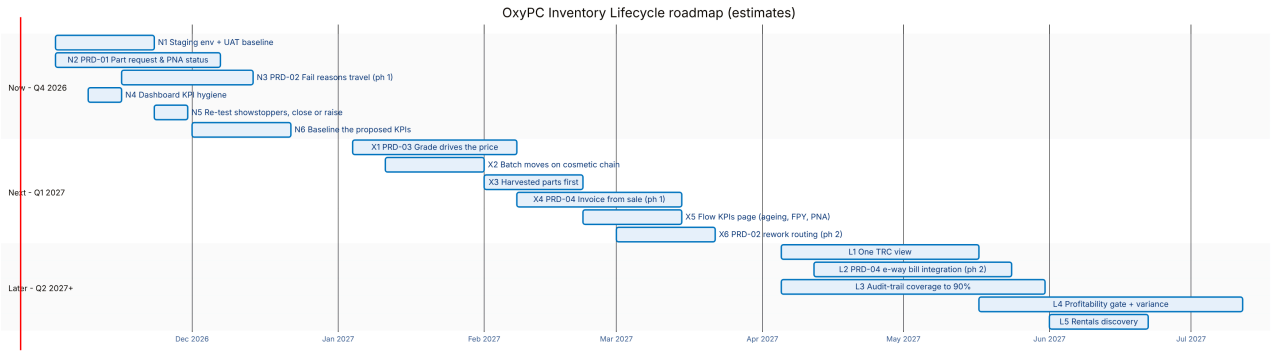
Roadmap

Horizons: **Now = Q4 2026** (Oct–Dec 2026) · **Next = Q1 2027** (Jan–Mar 2027) · **Later = Q2 2027 and beyond**. Built from the seven agreed improvement opportunities, `TASKS.md`, the open findings in `docs/audits/*.md`, the open gaps in `QA_UAT_REPORT.md` and the "Showstoppers" sheet of `docs/OxyPC_UAT_TestPlan.xlsx`. Effort is an **estimate**: S = up to 1 week, M = 2–4 weeks, L = 5+ weeks of one developer, including tests and UAT fixes.

1. Inputs and what they told us

Input	What it says today
Agreed opportunities	Part-request & PNA status · Fail reasons travel with the tag · Batch moves on the cosmetic chain · Grade drives the price · Harvested parts first · Invoice & e-way bill from the sale · One TRC view
`TASKS.md`	Every listed backlog item is ticked `[x]`. No open feature tasks remain in that file. "Current Sprint Focus" in `CLAUDE.md` says: next move to device/sale page enhancements.
QA defect log (`seed_qa_*.py`, QA Dashboard `/qa/defects`)	All seeded defects (BUG-001…BUG-213) are `Closed`. Live open defects must be read from the QA Dashboard on the target server before each planning cycle.
`QA_UAT_REPORT.md` (28 Mar 2026)	Still marked Open: G-08 WhatsApp "Not Connected" state, G-09 cosmetic routing check, G-11 dealer follow-up data, G-12 pagination on Telecalling and Market.
`docs/OxyPC_UAT_TestPlan.xlsx` (27 Apr 2026, Showstoppers)	P0 CSRF 403 expected / 307 seen; P1 no HTTPS; P1 backup automation unconfirmed; P2 default admin password; four route status mismatches (307 vs 302). Several look like test-harness redirects; they must be re-run on staging, not assumed fixed.
Audits (Apr 2026)	Open themes: audit trail ~35% of writes; no profitability gate / variance tracking; `expected_sale_value` never set (auto-scrap never fires); testing 3.0/10.
Code review for this document	Dashboard P&L From/To filter feeds only values the page no longer shows; `work_queue_devices` and `today_followups` are computed but not rendered; "active lots" counts all lots (`routers/dashboard.py::dashboard`).

2. Roadmap at a glance



3. Now — Q4 2026

Theme: **make blockers visible and keep the story on the tag.**

ID	Item	Outcome	Personas	Dependency	Effort (estimate)
N1	Staging environment + UAT baseline	A staging server that mirrors production, seeded with anonymised data, used for every UAT in 06. Today `deploy/oxypc-autodeploy.sh` deploys `origin/main` straight to production.	All; Admin	Server / VM; copy of prod DB with PII masked	S
N2	**PRD-01 Part Request & PNA Status**	Engineers see each request's status (Requested, Handed Over, Not In Stock, Sent for Sourcing, PNA) on their queue and get notified on handover. PNA links to the request.	Repair Engineer, Spare Parts Manager, Production Manager, TRC Manager	dbdiagram approval for `part_requests` ↔ `device_pna_parts` link	M
N3	**PRD-02 Fail Reasons phase 1** — keep every fail on the tag	A fail-history panel on Device Detail and on the receiving queue; stop clearing / overwriting `stress_notes`, `repair_notes`, `fqc_failure_reason` without a history row.	QC Inspector, Cosmetic Lead, Final QC Inspector, Repair Engineer	New `device_fail_events` table (dbdiagram approval)	M
N4	Dashboard KPI hygiene	Remove or wire the P&L From/To filter (it changes nothing visible today); fix "active lots" label (counts all lots); drop unused queries (`work_queue_devices` , `today_followups`); make "Revenue This Month" use the same override-aware source as "Profit This Month". See 05 §6.	Managers/Admin, Sales, Inventory Manager	KPI owners agree definitions	S
N5	Re-test 27-Apr showstoppers on staging	Each Showstopper row in `docs/OxyPC_UAT_TestPlan.xlsx` is either closed with evidence or raised as a defect in the QA Dashboard.	Admin, App Development Team	N1	S
N6	Baseline the proposed KPIs	Measured baselines for stage ageing, first-pass yield, PNA wait, parts fill rate, grade mix, sales cycle time (SQL in 05 §5), so Next-quarter targets are real.	Managers/Admin, TRC Manager	Read access to prod DB copy	S

4. Next — Q1 2027

Theme: **move faster, price smarter, bill from the sale.**

ID	Item	Outcome	Personas	Dependency	Effort (estimate)
X1	**PRD-03 Grade drives the price**	Final QC grade + device type + model give a suggested min/max selling price on Ready to Sale; Sales can override with a reason.	Final QC Inspector, Sales Executive, Managers/Admin	Single agreed grade scale (Vision Q5); price table approval	M (estimate)

ID	Item	Outcome	Personas	Dependency	Effort (estimate)
X2	Batch moves on the cosmetic chain	Cosmetic Lead selects many tags and advances them one stage in one action (today `routers/cosmetic.py::advance_stage` is one tag; `bulk_assign` only reassigns). Per-tag result alert.	Cosmetic Lead	None (no schema change expected)	M
X3	Harvested parts first	When a request is handed over, Part Master suggests matching Harvest stock before New stock; report the harvest share.	Spare Parts Manager, Repair Engineer	PRD-01 (request status)	M
X4	**PRD-04 phase 1** — invoice from the sale	One invoice per sale event (all tags of a multi-tag sale on one document), server-generated invoice number series, e-way bill number and date captured on the sale.	Sales Executive, Accounts	dbdiagram approval (`sale_invoices`); Accounts confirms numbering	M
X5	Flow KPIs page	Stage ageing, first-pass yield per QC stage, PNA wait time, parts fill rate, grade mix, sales cycle time on one management page using the Global Table.	Managers/Admin, TRC Manager, Production Manager	N6 baselines; N3 for clean fail data	M
X6	**PRD-02 phase 2** — rework routing	Stress and Cosmetic fails choose a reason category (as Final QC does) and route by rule; returned tags show "Rework #n".	QC Inspector, Cosmetic Lead, Final QC Inspector, Production Manager	N3	S

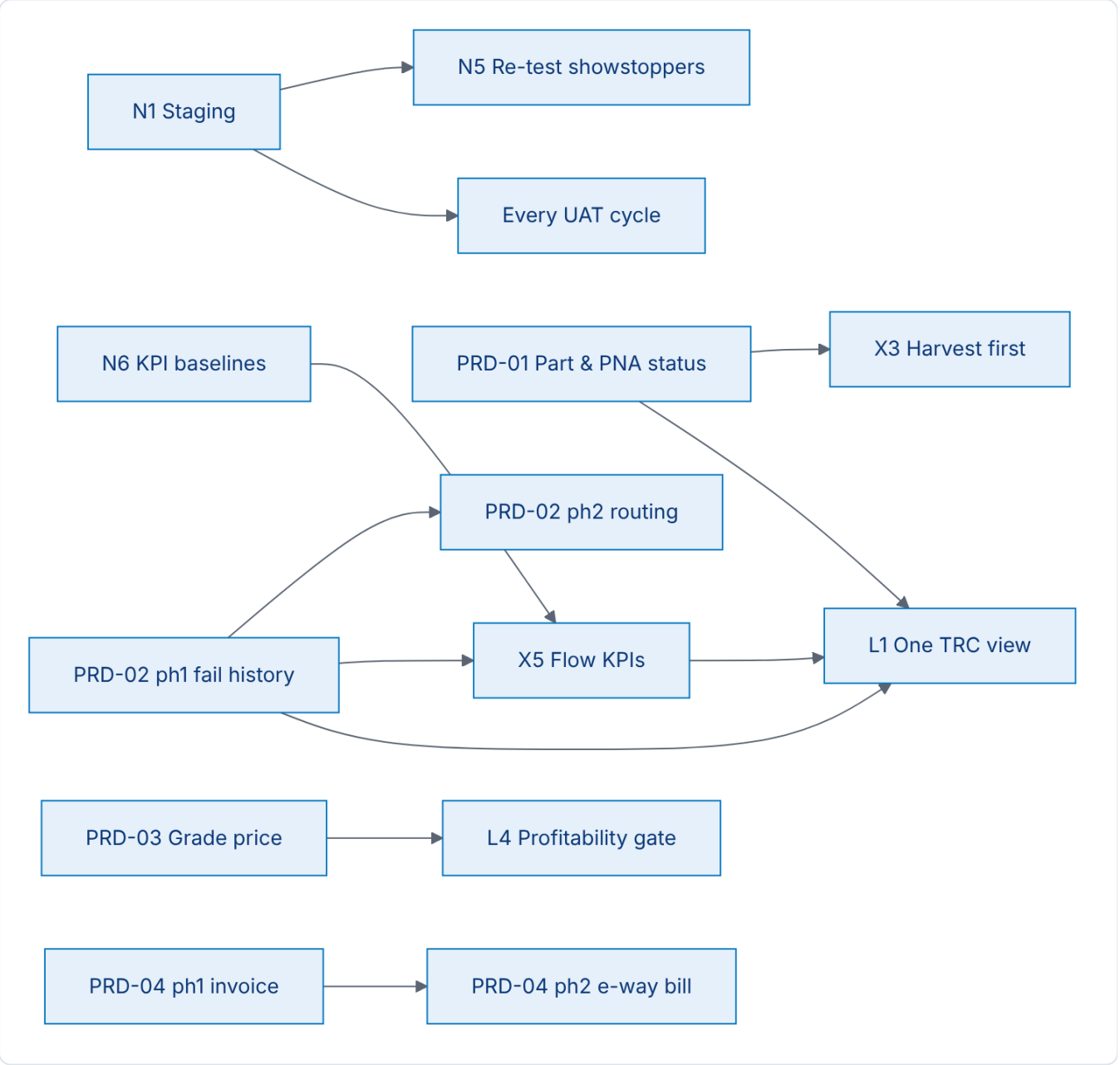
5. Later — Q2 2027 and beyond

Theme: **one view for the TRC and numbers you can defend.**

ID	Item	Outcome	Personas	Dependency	Effort (estimate)
L1	One TRC view	A single TRC Manager page: tags by stage and age, PNA list, open part requests, Final QC fails, today's daily-stock opening/closing, last physical audit result. Reuses existing queries.	TRC Manager, Production Manager	X5, PRD-01, PRD-02	L
L2	**PRD-04 phase 2** — e-way bill (and e-invoice if needed) via a GST provider	E-way bill generated from the sale, number written back, printed on challan and gate pass.	Accounts, Sales Executive	GSP account and credentials; Accounts decision on e-invoice (Vision Q4)	L
L3	Audit-trail coverage to ≥ 90%	IQC registration, user/role changes, spare-part movements, CRM deal edits all audited (gaps listed in `docs/audits/2026-04-29-claude-md-full-audit.md` §5).	Managers/Admin	None	L
L4	Profitability gate + variance tracking	Lot buy decision shows expected margin from grade mix; actual vs estimate per lot after sale (audit "Commercial 3.5/10").	Managers/Admin, Inventory Manager	X1 grade price table	L

ID	Item	Outcome	Personas	Dependency	Effort (estimate)
L5	Rentals discovery	Decide whether rentals belong in this app; if yes, write PRD. No rental code exists today.	Sales Executive, Accounts	Business input	S
L6	Remaining low gaps	G-08, G-11, G-12 from `QA_UAT_REPORT.md` (WhatsApp state, dealer follow-ups data, Telecalling/Market pagination).	Sales	None	S

6. Dependencies map



7. How we decide what moves

Rule	Detail
Blockers first	Items that stop a tag moving (parts, fails) beat items that make it look nicer.
Measure before target	A KPI target is set only after N6 baseline exists.
One schema review per quarter	Collect all schema asks for the quarter into one dbdiagram review (`CLAUDE.md` : no schema change without approval).
Release gate	Nothing ships to production without staging UAT sign-off for the affected personas (06).

8. Open questions

#	Question	Owner
1	Is there a live open-defect list beyond the seeded (closed) ones? Pull from `/qa/defects` on production before sprint planning.	QA lead
2	Who owns the GST provider account for e-way bill (L2)?	Accounts
3	Should batch cosmetic moves (X2) also allow "Skip Cosmetic → Final QC" in bulk? Single-tag skip exists in `advance_stage` .	Cosmetic Lead
4	Is the TRC view (L1) a new page or a tab on `/trc-production` ?	TRC Manager

PART D

KPI Definitions

Every figure below is read from the code on branch `staging-docs`. The formula column cites `file::function`. **Target** is "to be agreed" unless the business has already set one. **Owner** is the persona who answers for the number. Section 5 lists **proposed** KPIs that the app does not show yet.

1. Shared rules (apply to every dashboard figure unless stated)

Rule	Detail	Source
Trashed devices	Excluded: `Device.is_trashed == False`.	`routers/dashboard.py::dashboard`
Test entity	With no Entity filter, the External Partner test entity is excluded (NULL-safe).	`models/master.py::EXTERNAL_PARTNER_TEST_ENTITY`, `routers/dashboard.py::dashboard` (`_ent`)
Filters	Entity (multi), Device Type (multi, Master Data `device_type`), Zone (latest `DeviceLocationLog`, fallback `Device.location_id`).	`routers/dashboard.py::dashboard`
Refresh	Counts are cached for **30 seconds** when no filter is active (`_AGG_TTL = 30`). Filtered views run live. Page has no auto-refresh; figures update on reload.	`routers/dashboard.py` (`_AGG_CACHE`)
Who sees the page	Gated by the Module Permission matrix; every role that can open `/dashboard` sees the same analytics. Weekly Sales Price chart needs `can_view_pricing`.	`routers/dashboard.py`, `templates/dashboard.html`, `templates_config.py::_can_view_pricing`
Stage groups	**Cosmetic** = `cosmetic_received, cleaning, putty, dry_sanding, masking, painting, water_sanding, cosmetic_completed` (`models/device.py::COSMETIC_STAGES`). **Final QC** = `final_qc, final_qc_pass_hold, final_qc_fail_hold`. **L1/L2** = `l1, l2` (l2 legacy). **L3/L4** = `l3`. **Stress/QC** = `qc_check`. **Production** = `trc_production`.	`routers/dashboard.py` (`STEP_STAGES`)
Financial Year	Indian FY, April–March, picked by its start year (2025 = "2025-26").	`services/business_pl.py::fy_label`
Admin overrides	Business P&L month values can be overridden by admin; overrides flow into FY figures.	`routers/reports.py::save_business_pl_override`, `models/business_pl_override.py`

2. Overall Dashboard tiles (`/dashboard`)

Template: `templates/dashboard.html` and `templates/dashboard/_admin_analytics.html`. Computation: `routers/dashboard.py::dashboard` unless stated.

2.1 Device Stage Pipeline card (10 tiles)

Each tile counts devices passing the shared rules plus the stage clause below. Tiles link to the page whose count should match.

#	Tile	Definition	Formula / query logic	Owner	Target	Refresh
D1	IQC	Tags waiting for or in IQC	`count(Device) where current_stage = 'iqc'` (`PIPELINE_STEPS`)	IQC Inspector	to be agreed	30 s cache
D2	Stock In	Tags passed IQC, not yet allocated	`current_stage = 'stock_in'`	Inventory Manager	to be agreed	30 s
D3	Production	Tags with the Production Manager for allocation	`current_stage = 'trc_production'`	Production Manager	to be agreed	30 s
D4	L1/L2	Tags in L1/L2 repair	`current_stage IN ('l1','l2')`	Repair Engineer L1/L2	to be agreed	30 s
D5	L3/L4	Tags in L3/L4 repair	`current_stage = 'l3'` . Note: the linked page `/repair/l3l4` counts open `L3L4-%` WorkOrders and may differ (comment in `dashboard`).	Repair Engineer L3/L4	to be agreed	30 s
D6	Stress/QC	Tags in Stress Test	`current_stage = 'qc_check'`	QC Inspector	to be agreed	30 s
D7	Cosmetic	Tags anywhere on the paint line	`current_stage IN COSMETIC_STAGES`	Cosmetic Lead	to be agreed	30 s
D8	Final QC	Tags in Final QC incl. Pass/Fail Hold	`current_stage IN FINAL_QC_STAGES`	Final QC Inspector	to be agreed	30 s
D9	Ready to Sale	Sellable stock	`current_stage = 'ready_to_sale'`	Sales Executive	to be agreed	30 s
D10	Sold	All-time sold tags	`current_stage = 'sold'`	Sales Executive	n/a (cumulative)	30 s
D11	GRN (computed, not shown as a tile)	Tags mapped to a GRN	`grn_number IS NOT NULL AND grn_number <> '' AND is_active` (`GRN_WHERE`)	Inventory Manager	n/a	30 s
D12	Returned / Scrapped / Scrap for Sale (footer line)	Dead-end stage counts, shown apart from totals	`stage_counts['returned']` , `['scrapped']` , `['scrap_for_sale']` from `GROUP BY current_stage`	Inventory Manager	to be agreed	30 s
D13	Split Entity / Split Zone (toggles)	Same tile counts broken down by `Device.entity` or by Zone of resolved location	One grouped query: stage × entity × resolved location (`_pipe_count_by_entity` , `_zone_label`)	Managers/Admin	n/a	30 s

2.2 Financial Summary card (Financial Year)

All five figures come from the same functions as Business P&L, so the card always equals that report for the chosen FY.

#	Tile	Definition	Formula / query logic	Owner	Target	Refresh
F1	Total Investment	Device cost of tags **sold** in the FY	Σ `Device.device_price` joined to `Sale` by `sold_at` month, Apr–Mar, with Device COGS overrides — `services/business_pl.py::compute_year_device_cogs` via `compute_fy_totals`	Managers/Admin (Accounts)	to be agreed	30 s, keyed by FY
F2	Parts Spent	Parts changed on tags sold in the FY	Σ `PartRequest.qty_handed_over × SparePart.unit_price` where `PartRequest.status = 'received'`, joined to `Sale.sold_at` month; overrides applied — `services/business_pl.py::compute_year_parts_labour_cogs`	Spare Parts Manager	to be agreed	30 s
F3	Labour Spent	Flat labour per worked-on sold tag	`count(distinct Sale.device_id)` for sold tags with any `StageMovement.to_stage IN LABOUR_STAGES` (L1, L2, L3, Stress, all Cosmetic, all Final QC) × CostConfig `repair_labour_rate` (default ₹150); overrides applied — same function	Production Manager	to be agreed	30 s
F4	Total Revenue	Sales value in the FY	Σ `Sale.sale_price` by `sold_at` month, overrides applied — `services/business_pl.py::compute_year_revenue` via `compute_fy_revenue`	Sales Executive / Managers	to be agreed	30 s
F5	Net Profit	Gross profit for the FY	F4 − F1 − F2 − F3 (`fy_net_profit` in `dashboard`)	Managers/Admin	to be agreed	30 s

2.3 Category quick-view tiles (Laptops, Desktops, TFT Monitor, Tablet, Mini PC, Server)

#	Tile part	Definition	Formula / query logic	Owner	Target	Refresh
C1	Big number	Active stock of this category	Σ counts by `Device.sub_category` excluding stages `grn, sold, returned, scrapped, scrap_for_sale` (`CATEGORY_EXCLUDED_STAGES`)	Inventory Manager	to be agreed	30 s
C2	In Ready	Sellable units of this category	`category_counts[cat]['ready_to_sale']` (`laptops_available` etc.)	Sales Executive	to be agreed	30 s
C3	In TRC	Units still being worked on	`max(C1 − C2, 0)` (computed in template)	Production Manager	to be agreed	30 s

2.4 Summary stat cards

#	Tile	Definition	Formula / query logic	Owner	Target	Refresh
S1	Total Inventory	All active stock in the building	Σ `stage_counts` excluding `CATEGORY_EXCLUDED_STAGES` (`total_devices`)	Inventory Manager	to be agreed	30 s
S2	In Repair	Work-in-progress between Stock In and Ready	Pipeline D3 + D4 + D5 + D6 + D7 + D8 (`in_repair_count`)	Production Manager	to be agreed	30 s
S3	Revenue This Month	Sales value since the 1st of this month	Σ `Sale.sale_price` where `date(sold_at) >= first of month` (`_compute_financials`). **No overrides, not net of returns.**	Sales Executive	to be agreed	30 s

#	Tile	Definition	Formula / query logic	Owner	Target	Refresh
S4	Profit This Month	Gross profit this calendar month	Revenue − Device − Parts − Labour for the month, overrides applied — `services/business_pl.py::compute_month_profit`	Managers/Admin	to be agreed	30 s

2.5 Charts

#	Chart	Definition	Formula / query logic	Owner	Target	Refresh
G1	Stage Distribution (pie)	Work-in-progress mix	Pipeline counts for IQC, Stock In, Production, L1/L2, L3/L4, Stress, Cosmetic, Final QC (`admin_charts['stage_pie_*']`)	Production Manager	n/a	30 s
G2	Inventory by Category & Stage (bar)	Stock by category per pipeline step (Sold excluded)	`chart_groups` × `category_counts`	Inventory Manager	n/a	30 s
G3	Weekly Sourcing Price — Buyer PO / Seller PO	Value of dealer orders vs supplier POs per ISO week, last 8 weeks	Σ `DealerOrder.total_amount` by `order_date` ; Σ `CRMPurchaseOrder.total_amount` by `created_at` (`_weekly_series`)	Managers/Admin	n/a	30 s
G4	Weekly Sales Price — Ready to Sale / Sold (pricing-gated)	Tags entering Ready per week (count) vs sold value per week (₹)	count `StageMovement.to_stage = 'ready_to_sale'` ; Σ `Sale.sale_price` by `sold_at`	Sales Executive	n/a	30 s
G5	Weekly: Products in IQC / GRN / In Stock	Inflow per week	count `StageMovement.to_stage IN ('iqc','grn','stock_in')` by `moved_at`	IQC Inspector, Inventory Manager	n/a	30 s
G6	Weekly: Spare Parts — Sourcing Request / GRN / Harvest	Parts demand and inflow per week	count `PartSourcingRequest.created_at` ; count `PartsGRNLineItem` by `created_at` split on `is_harvest`	Spare Parts Manager	n/a	30 s

2.6 Role "My Work Queue" banners (top of page, by role)

#	Banner figure	Role(s)	Formula	Owner	Target	Refresh
Q1	devices awaiting IQC	`iqc_inspector`	`stage_counts['iqc']`	IQC Inspector	to be agreed	30 s
Q2	devices in L1 repair queue	`l1_engineer`	`stage_counts['l1']`	Repair Engineer	to be agreed	30 s
Q3	devices in L2 queue	`l2_engineer`	`stage_counts['l2']` (legacy stage)	Repair Engineer	n/a	30 s
Q4	devices in L3 queue	`l3_engineer`	`stage_counts['l3']`	Repair Engineer L3/L4	to be agreed	30 s
Q5	devices pending QC check	`qc_inspector`	`stage_counts['qc_check']`	QC Inspector	to be agreed	30 s

#	Banner figure	Role(s)	Formula	Owner	Target	Refresh
Q6	ready to sell / sold today / month revenue	`sales`, `sales_manager`, `telecaller`	`stage_counts['ready_to_sale']`; `count(Sale) where date(sold_at)=today`; Σ `sale_price` this month	Sales Executive	to be agreed	live (sold today) / 30 s
Q7	dealer outstanding / overdue orders	sales roles	Σ `DealerOrder.due_amount` where status in pending/confirmed/delivered; count where `due_amount > 0 AND payment_due_date < now`	Accounts	to be agreed	live
Q8	low stock alerts / parts used today / stock value	`spare_parts_manager`	count `SparePart.qty_in_stock <= min_stock_alert`; Σ `SparePartConsumption.qty_used` today; Σ `qty_in_stock × unit_price`	Spare Parts Manager	to be agreed	live
Q9	devices in Stock-In / "active lots"	`inventory_manager`	`stage_counts['stock_in']`; `count(Lot)` — **all lots, including trashed**	Inventory Manager	to be agreed	live

3. Key reports (`/reports/*`, `routers/reports.py`)

Router gate: `_REPORT_ROLES` (inventory_manager, qc_inspector, sales; admin and custom roles via matrix). P&L reports add `_require_financials` (inventory_manager, sales_manager). Receivables add `_require_receivables`.

#	Report / KPI	Definition	Formula / query logic	Owner	Target	Refresh
R1	Lot P&L — Revenue (net)	Sales value per lot net of customer returns	Σ `Sale.sale_price` per lot − Σ `coalesce(Return.refund_amount, Sale.sale_price)` for non-rejected returns — `routers/reports.py::lot_pl_report`	Managers/Admin	to be agreed	on load
R2	Lot P&L — Total Cost / Profit / Margin %	Whole-lot economics	Cost = `Lot.buying_price` + Σ `SparePartConsumption.total_cost` (by lot) + Σ `RepairAttempt.cost`; Profit = R1 − Cost; Margin = Profit / R1 × 100 — `lot_pl_report`	Managers/Admin	Margin ≥ `GradePriceMatrix.min_margin_pct` default 15% *(not enforced; to be agreed)*	on load
R3	Lot P&L — Cost / Profit on Sold Units	Fair profit for a part-sold lot	Sold buying cost (`device_price`, else lot price ÷ devices) + parts + labour on sold tags; Profit = R1 − that — `lot_pl_report`	Managers/Admin	to be agreed	on load
R4	Business P&L — Revenue, Device/Parts/Labour COGS, Gross Profit, Gross Margin %	FY by month	Monthly lists from `services/business_pl.py` (§2.2); GM % = (Revenue − COGS) / Revenue × 100 — `routers/reports.py::business_pl`	Managers/Admin, Accounts	to be agreed	on load
R5	Business P&L — Sales count / Avg Sale Price	Volume and price per FY	`count(Sale) where sold_at in FY`; Revenue ÷ count — `business_pl`	Sales Executive	to be agreed	on load

#	Report / KPI	Definition	Formula / query logic	Owner	Target	Refresh
R6	Business P&L — Inventory value	Cost of stock on hand	Σ `Device.device_price` where stage not in (sold, scrapped) — `business_pl`	Inventory Manager	to be agreed	on load
R7	Business P&L — Lots Sold	Lots fully resolved	count lots with ≥1 active device and none still in process (each active device sold, scrapped or replaced) — `business_pl`	Inventory Manager	to be agreed	on load
R8	Daily Stock — Opening / Closing / Net per stage	Stock at 00:00 of day vs 00:00 next day	Latest `StageMovement.to_stage` per device at the boundary (DISTINCT ON), excluding `sold` and `grn` stages and devices trashed before the boundary — `routers/reports.py::daily_stock_report`, `_stock_as_of`	TRC Manager, Accounts	Closing = Opening + In − Out (reconciles)	on load; files auto-generated daily 9 PM (`services/daily_stock_reports.py`)
R9	Daily Stock — Sold today	Tags in stock at opening that are sold by close	`_sold_in_day`	Accounts	to be agreed	on load
R10	Stock Aging	Units and cost by age bracket and stage	Age = today − `Device.created_at` (**device age, not stage age**); brackets 0–7, 8–30, 31–60, 61–90, 90+; cost = `device_price`; excludes sold/scrapped/returned — `routers/reports.py::stock_aging_report`	Inventory Manager	90+ days bucket: to be agreed	on load
R11	Overdue (stuck tags)	Tags not updated for N days	`Device.updated_at <= now − min_days` (default 3), stage not sold/scrapped — `routers/reports.py::overdue_report`	Production Manager	to be agreed	on load
R12	Receivables ageing	Dealer dues by days overdue	`DealerOrder.due_amount > 0`, status pending/confirmed/delivered, bucketed current / 1–30 / 31–60 / 61–90 / 90+ from `payment_due_date` — `routers/reports.py::receivables_report`	Accounts	90+ bucket: to be agreed	on load
R13	Sales Report	Sales list in range (default 90 days) incl. orphan "sold without Sale" rows	`Sale` joined to `Device`, `Lot`; plus `_sold_orphan_rows` — `routers/reports.py::sales_report`	Sales Executive, Accounts	Orphans = 0	on load
R14	Stage Movement log	Every stage move	`StageMovement` joined to `Device`, newest first — `routers/reports.py::stage_movement_report`	Managers/Admin	n/a	on load

4. Other operational tiles used as KPIs

#	KPI	Definition	Formula / query logic	Owner	Target	Refresh
O1	Production tiles: Tags at You / in L1/L2 / L3/L4 / Stress / Cosmetic / Final QC / Tags Returned	Work by stage for the Production Manager	Shared clauses in `routers/stock.py::_prod_tile_criteria` (active, not trashed); "Tags Returned" = `tag_return_status = 'Return for Repair'` in pipeline stages	Production Manager	to be agreed	on load
O2	PNA tags (Production)	L1/L2 tags with ≥1 active PNA part	`count(distinct Device)` join `DevicePNAPart.is_active` where stage in (l1, l2) — `routers/stock.py::trc_production_list`	Production Manager	to be agreed	on load
O3	Part Master: Below Min / Out of Stock / Stock Value / New vs Harvest	Parts health	`qty_in_stock <= min_stock_alert` ; live qty (stock − consumed − sold) ≤ 0; Σ `unit_price × qty_in_stock` ; count by `SparePart.source` — `routers/spare_parts.py::parts_list`	Spare Parts Manager	to be agreed	on load
O4	Open sourcing requests	Parts sent for procurement	`count(PartSourcingRequest) where status = 'open'` — `parts_list`	Spare Parts Manager	to be agreed	on load
O5	WorkID Status: total PNA	WorkIDs whose tag has active PNA	`routers/workid_status.py` (`total_pna`)	TRC Manager	to be agreed	on load

5. Proposed KPIs (not in the app yet)

All six are **proposed**. Data already exists in the tables named; no new capture is needed except where stated.

#	KPI (proposed)	Definition	Formula / query logic (proposed)	Owner	Target	Refresh
P1	Stage ageing	Days a tag has been in its **current** stage; median and count > 7 / 30 days per stage	For each device, latest `StageMovement` with `to_stage = Device.current_stage` and `exited_at IS NULL` : `now − moved_at` . Fallback `Device.updated_at` where no row. Group by stage.	Production Manager, TRC Manager	baseline: to be measured; then to be agreed	daily (9 PM with Daily Report)
P2	First-pass yield (FPY) per QC stage	Share of tags that pass a QC stage at the first attempt	**Stress:** tags whose first exit from `qc_check` goes forward (to a cosmetic stage) ÷ all tags that exited `qc_check` in period. **Final QC:** tags whose first decision is `final_qc_pass_hold` ÷ tags with any Final QC decision. From ordered `StageMovement` rows per device.	QC Inspector, Final QC Inspector	baseline: to be measured; proposed ≥ 85% at Final QC	daily

#	KPI (proposed)	Definition	Formula / query logic (proposed)	Owner	Target	Refresh
P3	PNA wait time	Time a part stays PNA on a tag	`DevicePNAPart.cleared_at − marked_at` for cleared rows (median, P90); open rows aged `now − marked_at`. Note: re-marking reuses the row, so only the latest cycle is kept (`models/pna_part.py`).	Spare Parts Manager	baseline: to be measured; proposed median ≤ 5 working days	daily
P4	Parts fill rate	Share of part requests handed over within 1 working day	`PartRequest` with `status IN ('handed_over','received')` and `actioned_at − created_at ≤ 1 working day` ÷ all requests raised in period (excl. cancelled). Caveat: `actioned_at` is overwritten by the later Verify step (`validate_receiving`) — PRD-01 adds a dedicated `handed_over_at`.	Spare Parts Manager	baseline: to be measured; proposed ≥ 85%	daily
P5	Grade mix	Share of Final-QC-passed tags by grade, by lot and month	count `Device.grade` for tags with a `StageMovement.to_stage = 'final_qc_pass_hold'` in period ÷ total	Final QC Inspector, Managers/Admin	to be agreed	daily
P6	Sales cycle time	Days from Ready to Sale to Sold	`Sale.sold_at − (first StageMovement.to_stage = 'ready_to_sale' for the device)`; median by category and grade	Sales Executive	baseline: to be measured; then to be agreed	daily

UX (proposed): one "Flow KPIs" page, summary tiles on top, one Global Table per KPI with drill-down to tags (Roadmap X5).

6. Findings to fix (from reading the code)

#	Finding	Where	Suggested fix	Roadmap
K1	P&L From/To filter changes only values the page no longer shows (`total_revenue`, `total_investment`, `overall_profit`, `lot_pl`). The Lot P&L table macro `lot_pl_table()` is defined but never rendered.	`routers/dashboard.py`, `templates/dashboard.html`	Remove the filter or render the table	N4
K2	"Revenue This Month" ignores admin overrides; "Profit This Month" uses them. The two can disagree.	`_compute_financials` vs `compute_month_profit`	Use `compute_year_revenue(...)[month]` for S3	N4
K3	"active lots" counts every lot, including trashed.	`user_queue['lot_count']`	Filter `Lot.is_trashed.isnot(True)` or relabel	N4
K4	`work_queue_devices` and `today_followups` are queried on every load but not displayed.	`dashboard`	Remove or render	N4
K5	Stock Aging measures device age from creation, not time in stage.	`stock_aging_report`	Add P1 Stage ageing; keep R10 as "device age"	X5

#	Finding	Where	Suggested fix	Roadmap
K6	Dashboard Lot P&L (`_compute_lot_pl`) and `/reports/lot-pl` use different cost rules (dashboard adds cosmetic rate × cleaning entries and labour fallback; report nets returns).	both	Retire the dashboard copy (K1) or share one function	N4

7. Sign-off of definitions

KPI group	Owner to sign	Signed (name / date)
Pipeline and stock (D, C, S1–S2, R6–R11, O1–O2)	Inventory Manager + Production Manager	
Financial (F, S3–S4, R1–R5, R12)	Managers/Admin + Accounts	
Parts (F2, Q8, O3–O4, P3–P4)	Spare Parts Manager	
Quality (P2, P5)	Final QC Inspector + QC Inspector	
Sales (Q6–Q7, R13, P6)	Sales Executive lead	
TRC (O5, P1)	TRC Manager	